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CityGaze: A Human-AI Gaze-Informed Workflow for Context-Aware Social Sharing on OHMDs

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Abstract

CityGaze: A Human-AI Gaze-Informed Workflow for Context-Aware Social Sharing on OHMDs

by

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Current head-mounted displays often rely on cumbersome manual input for social sharing, creating a bottleneck for situated content creation and spatial engagement. We introduce *CityGaze*, a context-aware platform for the asynchronous sharing of location-tagged posts termed *Glimpses*. Addressing the gap in low-friction interaction, *CityGaze* utilizes a "Look-and-Speak" paradigm that integrates gaze cues with speech for fluid content creation without requiring manual menus. An AI-assisted workflow extracts visual highlights via gaze-driven segmentation and co-authors captions. For consumption, *CityGaze* employs a two-stage delivery mechanism using proximity and visual-semantic matching to proactively surface relevant posts. An exploratory study (N=12) demonstrated high usability and low cognitive load. Findings reveal that *CityGaze* promotes spontaneous sharing and spatial discovery while effectively balancing attention between the physical environment and the digital social layer. This work provides insights into the design of proactive, multimodal interfaces that support digital co-presence in social environments.

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Chapter 1

Introduction

Social media platforms have become deeply embedded in our everyday routines, connecting people to a wealth of information, ranging from restaurant reviews and local events to personal stories and recommendations. Unlike static factual details such as addresses and operating hours, user-generated content on platforms like Facebook and Instagram reflects diverse personal experiences and subjective perspectives. This rich, constantly updated content not only provides practical information but also potentially inspires deeper exploration and understanding of specific places through the eyes of others [44, 20, 13].

1.1 Problem Definition and Scope

This work targets the problem of enabling on-the-go creation and consumption of context-aware social media content without forcing users to switch attention to a handheld device. Optical See-Through Head-Mounted Displays (OST-HMDs, or OHMDs) represent a promising technology for seamlessly integrating digital content into the user’s real-world view by overlaying visual information directly onto transparent lenses. This heads-up computing paradigm [66] reduces the need to physically switch between devices, enabling more fluid and efficient multitasking in everyday life [8, 9]. Recent advancements have made OHMDs lighter, more ergonomic, and increasingly wearable, fostering innovative ways to blend digital and physical environments, particularly for activities such as additional information acquisition [14, 9] and content recording [8]. Collectively, these developments underscore the potential of OHMDs as next-generation interfaces that naturally

augment and complement human activity.

1.2 Problem Significance and Motivation

However, most existing commercial OHMDs primarily serve as external displays, either extending content from traditional desktops or simply integrating context-aware conversational artificial intelligence (AI) for question answering. These devices do not fundamentally change the user interface paradigm established by mobile or desktop devices. Instead, they transpose it onto the glasses directly. When using the extended screen, we still see the user interfaces of the phone and computer, rather than a dedicated interface designed for OHMDs. As a result, it remains unclear how OHMDs can facilitate the immediate and seamless sharing and access of user-generated content on social platforms, where the primary appeal is the ability to share and glance at interesting moments instantly and with limited interruption.

While prior work on OHMDs has focused mainly on documentation and context queries for travel or daily tasks [8, 9], these approaches have offered limited support for social interaction. In practice, people now rely heavily on social media, not only at home, but also while on the go. For example, users often check food review platforms before ordering the meal at the restaurant, or browse Facebook in front of shopping malls for promotions and local news. Social media has become an integral part of daily life, closely tied to spontaneous decision-making and sharing experiences in various public settings, a context that prior applications in OHMD settings have not supported well. Therefore, instead of focusing on event documentation or question-answering, this work aims to enable seamless, real-time social sharing and browsing directly from OHMDs, bringing the immediacy and interactivity of social media into users' everyday environments.

Beyond these interface limitations, prior work on OHMDs has focused mainly on documentation [8] and context queries [9] for travel or daily tasks, offering limited support for social interaction and content sharing. Yet in practice, people rely heavily on social media, and its use is ubiquitous beyond the home [26]: checking

restaurant reviews before ordering a meal, browsing promotions near a shopping mall, or sharing moments from daily outings. Social media is now closely tied to spontaneous, location-dependent decision-making in public settings [64], a use case that existing OHMD applications have largely overlooked. This work therefore shifts the focus from event documentation and question-answering to enabling seamless, real-time social sharing and browsing directly on OHMDs, bridging the gap between the immediacy of social media and the heads-up nature of wearable displays.

1.3 Project Aims and Objectives

In this work, I introduce *CityGaze*, a real-time, context-aware social media platform designed for on-the-go use with OHMDs. *CityGaze* enables users to create, share, and browse lightweight, context-tagged social media posts, referred to as *Glimpses*, with limited interruption while on the move.

To create a *Glimpse*, the user captures a first-person view (FPV) along with their gaze position, and collaborates with an AI assistant to compose a concise, informative caption through a human-AI co-creation workflow. Published *Glimpses* are proactively surfaced to nearby community members based on relevance and location context. To consume content, users can select any *Glimpse* via a ring-mouse controller to inspect further details or leave social reactions.

1.4 Thesis Overview

This thesis is organized as follows. Chapter 2 reviews related work on context-aware social sharing, social media in extended reality, and visual representations for low-interruption sense-making on OHMDs. Chapter 3 provides an overview of the two-phase study design and the ethical considerations common to both studies. Chapter 4 reports a formative study that evaluates the proposed gaze-driven co-creation workflow and the *Glimpse* representation to inform design iterations for exploratory deployment. Chapter 5 presents the full *CityGaze* system, including a walkthrough of the end-to-end interaction flow and the system architecture.

CHAPTER 1. INTRODUCTION

Chapter 6 describes an exploratory study conducted in real-world travel settings to examine on-the-go use in situ. Chapter 7 reports the study results, covering usability metrics and qualitative findings. Chapter 8 discusses the implications of the findings, and Chapter 9 concludes the thesis.

This thesis makes the following contributions:

1. The *CityGaze* system, a context-aware social media platform designed for OHMDs that supports both content creation and consumption in on-the-go settings. Unlike prior OHMD applications that focus on documentation or question-answering, *CityGaze* enables real-time social sharing and browsing with limited interruption to ambulatory activities.
2. The *Glimpse* representation and a "Look-and-Speak" interaction paradigm for low-effort, on-the-go content creation. Users direct their gaze at an object of interest and speak a free-form comment. An AI-assisted co-creation pipeline then refines the spoken input and extracts a visual highlight via gaze-driven segmentation, producing a lightweight, context-tagged social media post.
3. A two-stage context-aware content delivery mechanism that surfaces relevant posts based on geographic proximity and real-time visual-semantic matching, offering an alternative to engagement-driven algorithmic feeds by grounding content relevance in the user's immediate physical surroundings.
4. Empirical findings from two user studies: (a) a formative study that validates the co-creation workflow and informs design iterations through ablation analysis and retrospective think-aloud sessions, and (b) an exploratory study that examines usage in multiple real-world contexts, yielding insights into how context-aware social media on OHMDs can enhance place perception and social connectedness without compromising situational awareness.

Chapter 2

Literature Review

In this chapter, I begin by reviewing the role of context-awareness in social sharing and the existing methods for social content presentation strategies. I then discuss the evolution of social media platforms into extended reality and the unique design challenges of supporting on-the-go use. Finally, I explore the use of visual content for sense-making, highlighting object-based approaches and their implications for minimizing interruption in OHMD-based social media platforms.

2.1 Context-Aware Social Content Sharing

Context-awareness is deeply intertwined with social media. Major platforms such as Facebook and Instagram personalize feeds using user profiles, inferred interests, and post metadata. A particularly important form of metadata is geolocation. Prior systems have investigated how to interpret and leverage geotags to organize and visualize place-linked social media on maps (e.g., NewsStand [58] and TwitterStand [54]). These projects primarily support place-based browsing through map-centric interfaces. Beyond 2D maps, other work has pursued richer spatial immersion by mapping 2D content onto three-dimensional (3D) mixed-reality worlds [16, 15] and by reconstructing 3D representations of the world [1, 53, 56, 59, 63].

In addition to location, the user’s egocentric view can provide valuable cues for interpreting intent and delivering timely assistance. Recent systems have explored personalized AI assistants in various situated settings [11, 19]. Within heads-up computing, prior work has studied knowledge acquisition [37, 9] and travel experience

documentation [8] from a single-user perspective. However, these efforts largely focus on individual workflows and do not examine how egocentric, in-situ context can be used to support asynchronous social interaction, where community members contribute, filter, and revisit shared content tied to the same physical places.

Together, these findings suggest that combining egocentric, in-situ context with community-generated content could enable a new form of asynchronous, place-centered social interaction, one that goes beyond map-based visualization or single-user assistance. Yet how to design such a system, recording context-tagged social posts, filtering them by the viewer’s real-time surroundings, and surfacing relevant community contributions on the move, remains largely underexplored.

2.2 Social Interaction on Extended Reality

Social media platforms are deeply embedded in everyday life. People use them to consume news, stay connected with family and friends [30], and participate in discussions within shared-interest online communities [65, 42].

At the same time, exploration of the digital world has expanded beyond desktop, laptop, and mobile phone screens. In recent years, wearable devices, especially extended reality (XR) headsets, have enabled users to carry social activities into immersive and spatial computing environments. In synchronous communication, for example, Blended Whiteboard introduces complementary affordances between physical and virtual whiteboards to support effective collaboration [21]. Thing2Reality leverages a 2D-to-3D artifact workflow to support remote social interaction [29]. Spacetime enables powerful individual work and fluid collaboration in virtual reality (VR) [61]. In asynchronous communication, Du et al. and Kukka et al. anchor social media posts in the 3D world [16, 15, 33]. Beyond fully immersive settings, BISHARE demonstrates that smartphones and head-mounted AR can serve complementary roles through bidirectional interaction, with each form factor contributing distinct affordances that the other cannot easily replicate [67].

Inspired by Du et al. [16, 15] and Kukka et al. [33], as well as emerging use

cases for augmented reality (AR) glasses, I am interested in bringing location-aware social media into everyday life. However, rather than spatially anchoring posts in 3D space as prior work does, lightweight smart glasses call for a different approach: casual, heads-up social interactions that support social connection without fully immersing users in a purely digital world, a promising yet largely underexplored direction. Unlike immersive VR environments, everyday smart glasses impose tighter constraints on visual occlusion, interaction effort, and situational awareness. How to design social media that respects these constraints while still enabling seamless on-the-go sharing and browsing remains an open question.

2.3 Enhancing Sense-Making with Visual Elements

Visual content has been widely adopted to complement text-only material, enriching information transmission and enhancing the accuracy of intent, which contributes to content sense-making. Prior work has largely explored adding visual to aid in audio content sense-making. For example, Crosscast was proposed to add visuals to audio travel podcasts, allowing travelers to better understand the places discussed in the audio [62]. Similarly, AudioXtend was developed to augment audiobook experiences during daily tasks by integrating AI-generated visuals through OHMDs, thus enhancing efficiency for multitaskers [57]. Beyond single-person use cases, researchers have also explored ways to integrate visuals into human conversations. For example, Visual Captions is proposed to be empowered by a large language model (LLM) to suggest relevant visuals proactively during open-vocabulary conversations [41].

Among the various visualization patterns explored, object-based visual elements, which isolate, extract, or emphasize specific regions of interest, have proven particularly effective for reducing visual noise and directing attention. Prior work leverages this principle along two complementary directions. The first is object extraction for content sharing: ThingShare enables users to create digital copies of physical objects from video feeds for sharing in remote meetings [28]; ThingMoji supports visual

CHAPTER 2. LITERATURE REVIEW

content referencing and precise intent anchoring in one-to-many communication [30]; Thing2Reality further extends object extraction into XR environments, allowing users to materialize physical objects as 3D renderings for remote discussion [29]; VideoSticker automates object detection and tracking in instructional videos, linking extracted stickers to transcripts so that learners can rapidly locate moments of interest [10]. The second is selective highlighting to manage attention: iBall, for example, highlights basketball players and dims the surrounding background based on the viewer’s gaze, helping casual fans follow the action amid visual clutter [68]. Across these diverse application contexts, the common insight is that isolating or emphasizing task-relevant objects from their surroundings can substantially improve content comprehension while lowering cognitive load.

Although prior practices show great potential for using visual augmentation to enhance sense-making, including both scene-based and object-based approaches [29], the design of thumbnails for social media posts, specifically to minimize interruption on OHMDs, remains largely underexplored. On the one hand, large, complete visual content can enhance users’ understanding of images but also increases visual occlusion and may interfere with perception of the real world. On the other hand, smaller visual elements, such as cut-outs and stickers, reduce this impact but may sacrifice information. Balancing informativeness with limited visual interruption is therefore a key challenge in designing thumbnails for OHMD-based social media posts. Prior work on object-based representations, including cut-outs [30, 28], stickers [10], and gaze-driven highlights [68], suggests that isolating salient visual elements can reduce noise while preserving key information, making these approaches promising candidates for compact, low-interruption content design on OHMDs.

Chapter 3

Study Overview

This research began with a formative study to (1) provide an initial evaluation of the *Glimpse* design and (2) elicit design considerations to guide subsequent heads-up deployment. Building on these findings, I iteratively designed and implemented the end-to-end system, *CityGaze*. I then conducted an exploratory user study in real-world travel settings to examine on-the-go use in situ. Prior to participation, all participants provided informed consent and agreed to the anonymous collection of study data, and all data were analyzed and reported in a manner that preserved participant anonymity. Participants were compensated at a locally standard rate upon completing the study.

Chapter 4

Formative Study

To evaluate the proposed gaze-driven co-creation workflow and refine *CityGaze* for real-world, heads-up use, I conducted a formative study with two complementary objectives: (1) verification of effects on content quality and (2) elicitation of design requirements for exploratory deployment. At this stage, the goal was not an end-to-end field deployment. Instead, the study focused on two core *Glimpse* components that later formed the system backbone: (1) a compact visual highlight and (2) a concise human-AI co-created caption. The study comprised two parts: Part 1 (Creator-Side Probe) and Part 2 (Viewer-Side Probe). Part 1 gathered input from real creators to inform the ablation design in Part 2. Part 2 conducted an ablation study to isolate the contributions of visual highlighting and text co-creation to perceived post quality, followed by retrospective think-aloud sessions to derive design implications for heads-up, on-the-go use.

4.1 Part 1: Creator-Side Probe

4.1.1 Method

I recruited 5 engaged content creators (E1-E5; 2 female, 3 male; ages 22-28; posting frequency: 5-12 visually dominated posts per month) with diverse backgrounds such as tourism, science communication, HCI research, and engineering. To improve efficiency and provide adequate coverage of scenarios, I simulated gaze interaction using a mouse-controlled cursor and used pre-captured first-person view (FPV) photos as prompts. Each creator completed 20 image tasks (100 image sources total)

and produced one caption under each of two paired conditions that shared the same gaze input. The scenarios covered include cultural and heritage exhibitions ($N=11$), dining places ($N=19$), parks ($N=21$), stores ($N=35$), and street views ($N=14$). After each paired trial, creators rated perceived **intent alignment**, the degree to which the resulting text matched what they intended to express, on a 7-point Likert scale (1 = strongly misaligned, 7 = strongly aligned). This procedure yielded 100 paired observations.

4.1.2 Results

Figure 4.1 shows the distribution of creator-side intent-alignment ratings. A Wilcoxon signed-rank test indicated a significant difference between conditions ($W = 213.5$, $p < .001$). Ratings in the Co-Creation condition ($Mean = 6.03$, $SD = 1.51$, Median = 7.0) were concentrated at higher scores (6-7), whereas ratings in the AI-Only condition ($Mean = 3.20$, $SD = 2.07$, Median = 3.0) were more dispersed toward lower scores (1-4), suggesting that co-creation better preserves creator intent.

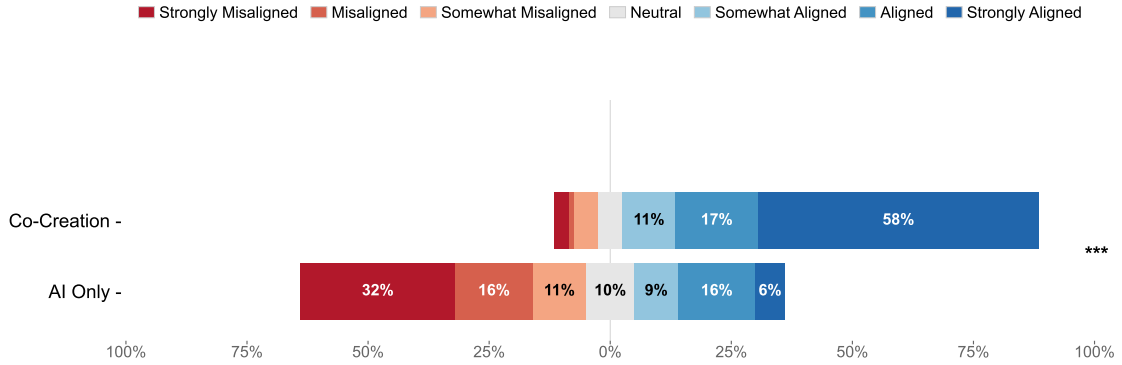


Figure 4.1: Formative study Part 1: Creator-side results (intent alignment). Diverging stacked distributions of intent-alignment ratings for human-authored captions with AI refinement (Co-Creation) and AI-only captions (AI-Only). Asterisks denote significance from a Wilcoxon signed-rank test ($*** : p < .001$).

4.2 Part 2: Viewer-Side Probe

4.2.1 Ablation Design

To disentangle component effects, I used a 2×2 ablation over two factors: **H** (compact visual highlight) and **C** (co-created caption based on a user comment).

- **C1 (Glimpse; H+C)**: Highlight + co-created caption (user comment + AI refinement)
- **C2 (w/o H; C only)**: No highlight (full image) + co-created caption
- **C3 (w/o C; H only)**: Highlight + AI-only caption (no user comment)
- **C4 (w/o HC; neither)**: No highlight (full image) + AI-only caption

12 raters (V1-V12; 6 female, 6 male; ages 21-53; 3-35 hours/week browsing visually dominated content) evaluated the generated posts. Each rater completed 100 tasks. Each task contained the 4 condition variants derived from the same source moment. Posts were presented in randomized order with anonymous IDs (i.e., Content 1-4), following a similar procedure used in prior work [9]. To discourage inattentive ratings, I enforced a minimum of 30 seconds before a rater could submit ratings for each moment. In total, this yielded $12 \times 100 = 1200$ moment-level trials and 1200 ratings per condition (i.e., 4800 condition instances overall). Raters used a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree) to assess the 4 metrics listed in Table 4.1. These measures were informed by prior literature (**Q1** [3, 51], **Q2** [39], **Q3** [50, 22, 46], **Q4** [35, 12]) and pilot testing.

4.2.2 Results

As shown in Figure 4.2, conditions with highlighting (**C1** and **C3**) consistently outperformed those without highlighting (**C2** and **C4**). Overall, these results suggest that **visual highlighting (H)** is the primary driver of perceived quality. In contrast, the viewer-side ratings did not show a large difference between conditions

Table 4.1: Viewer-side evaluation metrics in the formative study.

Metric	Definition
Q1: Gist	I can quickly understand what the creator intended to talk about.
Q2: Coherence	The image and text match well and make sense together.
Q3: Appeal	This post catches my eye and holds my interest.
Q4: Exploration	This post makes me want to go check it out in person.

with and without a **user comment (C)**. However, the Part 1 results indicate that incorporating a user comment (i.e., co-creation) better aligns the resulting caption with what creators intended to share, improving faithfulness to creator intent without degrading performance on the 4 viewer-side metrics.

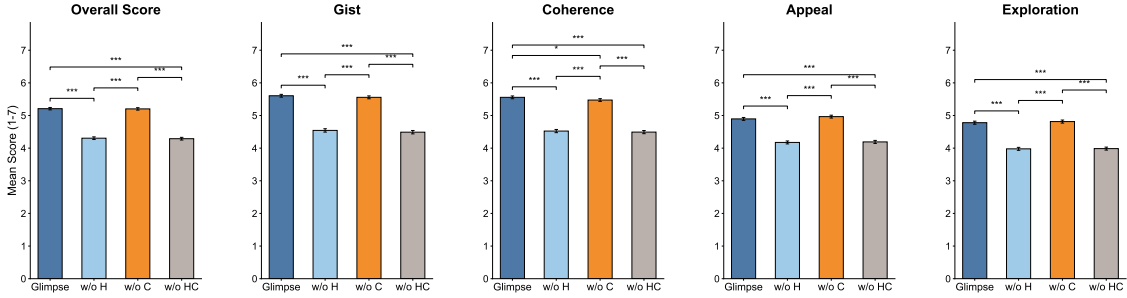


Figure 4.2: Formative study Part 2: Ablation results. Mean ratings for Overall, Gist, Coherence, Appeal, and Exploration across four conditions (C1: Glimpse; C2: w/o H; C3: w/o C; C4: w/o HC). Asterisks denote Wilcoxon-test significance (* : $p < .05$; *** : $p < .001$).

4.3 Design Considerations

Beyond rating differences, viewers’ retrospective think-aloud data surfaced constraints of heads-up, on-the-go viewing that shape how *Glimpse* should be visually presented. Using inductive thematic analysis, I synthesized these findings into five display-oriented design considerations (**D1–D5**), summarized in Table 4.2.

CHAPTER 4. FORMATIVE STUDY

Table 4.2: Five design considerations derived from the formative study.

ID	Guideline	Evidence
D1	Scene relevance: prioritize feed items aligned with the user’s current physical scene and suppress irrelevant content.	V1, V2, V3, V4, V5, V6, V9, V10, V11, V12; [36, 40, 9]
D2	Proximity-aware feed: surface nearby, geotagged items to strengthen coupling between <i>Glimpse</i> content and the surrounding context.	V1, V8, V10, V11, V12; [16, 15]
D3	Glanceable, low-interruption display: support brief, rapid readability while preserving situational awareness and the user’s primary physical activity.	V1, V2, V3, V4, V6, V8, V10; [45, 66, 31, 32]
D4	Proactive delivery: trigger content contextually to reduce explicit user input and manual browsing.	V2, V4, V6, V8, V10, V12; [36]
D5	Lightweight social reactions: provide quick, low-effort social feedback through condensed visual cues, enabling at-a-glance, heads-up awareness without opening detailed views.	V1, V4, V5, V6, V8, V10, V12; [55, 17]

Together, these findings informed the end-to-end *CityGaze* design and implementation in Chapter 5.

Chapter 5

CityGaze System

5.1 System Overview

CityGaze enables users to effortlessly share and discover location-anchored content based on their physical surroundings, enriching on-the-move activities with user-generated content and online social interaction. *CityGaze* emphasizes low-effort authoring and glanceable consumption: users create a *Glimpse* through natural gaze combined with a voice comment, and browse nearby *Glimpses* in a compact peripheral display that reduces visual occlusion and interaction overhead in ambulatory activities.

The system pipeline is structured around two core workflows: *Glimpse* creation and *Glimpse* consumption (Figure 5.1). During creation, the system captures a first-person view (FPV) photo together with the user’s gaze point, performs gaze-driven image segmentation to extract the gazed-at object as a visual sticker, and uses a multimodal large language model (LLM) to simultaneously refine the user’s spoken comment and select the most appropriate segmentation result. During consumption, a two-stage context-aware filtering pipeline, combining geospatial proximity with visual-semantic relevance, continuously surfaces the most contextually appropriate *Glimpses* as the user moves through the environment. A ring mouse controller mediates interactions to minimize manual interference, allowing users to remain engaged in ambulatory activities. Details regarding the system’s prompt design are provided in Appendix C.

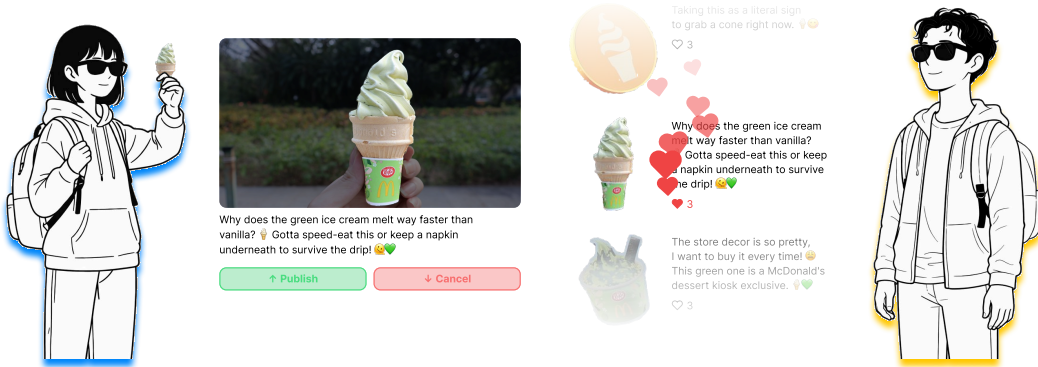


Figure 5.1: Concept of the *CityGaze* use case. Left: Alice looks at an ice-cream cone, activates capture, adds a spoken comment, and previews the generated *Glimpse* before publishing. Right: Bob, a nearby viewer, sees the published *Glimpse* in a glanceable feed rendered on the peripheral region of the OHMD and gives a lightweight "like" social reaction.

5.2 *Glimpse* Creation

The creation workflow allows a user to produce a *Glimpse* in a single continuous interaction: the user long-presses the right button on the ring mouse controller to initiate capture, speaks while holding the button, and releases to finalize. The pipeline is designed so that image segmentation and voice recording proceed in parallel, reducing total creation latency (Figure 5.2).



Figure 5.2: The *Glimpse* creation workflow. Left: The user initiates capture by long-pressing the ring mouse button. The system instantly captures the first-person view frame and gaze point, and records the user's spoken comment until the button is released. Center: A preview of the generated *Glimpse*, comprising the AI-selected cutout sticker highlighted against the original scene and the refined caption, is displayed with publish (↑) and cancel (↓) options. Right: The confirmed *Glimpse* is published and becomes available for nearby viewers to browse and react to.

5.2.1 Step 1: Gaze-Augmented Photo Capture

The moment the right button is pressed down, the system captures a single frame from the OHMD’s world camera via the Pupil Core eye tracker and simultaneously records the user’s gaze coordinates. The raw 2D gaze position provided as normalized coordinates by the eye tracker is converted to pixel coordinates on the captured frame. A gaze-overlay image, the original frame annotated with a semi-transparent red dot (radius = 20 px, $\alpha = 0.5$) at the gaze point, is also generated and stored as visual context for downstream LLM calls.

5.2.2 Step 2: Gaze-Driven Image Segmentation

The captured frame and gaze-point pixel coordinates are forwarded to the Segment Anything Model (SAM), which uses the gaze point as a single positive point prompt with multi-mask output enabled. SAM returns three masks at different granularity levels. These are sorted by mask area in ascending order to yield a small, medium, and large mask (indices 0, 1, 2). For each mask, two visual outputs are generated: (a) an RGBA cutout, in which the masked region is cropped from the frame with a 10 px padding and all pixels outside the mask are set to transparent; and (b) a highlight image, a full-frame composite where the background is darkened to 50 % of its original brightness while the masked region retains full brightness, providing a preview of the segmentation extent. These 6 artifacts (3 cutouts and 3 highlights) are stored in a temporary folder for preview and downstream selection.

5.2.3 Step 3: Concurrent Voice Input

While the segmentation executes on the back end, the user continues to hold the right button and speaks a free-form comment on what they are looking at. Speech is transcribed in real time on the client, which streams a running transcript back to the interface.

5.2.4 Step 4: Parallel AI Finalization

When the user releases the right button, the raw transcript is sent to the back end along with the capture session identifier. Two multimodal LLM calls are dispatched concurrently in a thread pool:

- **Mask Selection** The LLM receives the original full frame, the three cutout images, and the user’s spoken text. It is instructed to select the cutout index (0, 1, or 2) that best matches the user’s spoken intent, balancing semantic completeness (the cutout must contain the entire referred object), cleanliness (no irrelevant background), and visual integrity (no essential parts cropped off). The model returns a JSON object containing the selected index.
- **Speech Refinement** The LLM receives the gaze-overlay image and the raw transcript. It is prompted to refine the transcript into a concise (≤ 200 characters), social-media-style caption: removing filler words (e.g., "uh", "um"), correcting misheard words using the visual context indicated by the red gaze dot, and adding appropriate emoji. The output is the refined caption text only.

5.2.5 Step 5: Preview and Publish

The selected cutout and refined caption are returned to the client and displayed as a preview. The user presses the up button on the ring mouse to confirm publication, or the down button to discard. Upon confirmation, a third LLM call classifies the published post: given the refined caption and the selected cutout image, the model assigns a single-word content category (e.g., "food", "nature", "architecture") and 3-6 descriptive keyword tags (e.g., ["egg waffle", "snack", "street food"]). The complete post record, comprising a unique identifier, the refined caption, URLs to the cutout, the full FPV photo, and the highlight image, geographic coordinates from the GNSS receiver, a Unix timestamp, and the assigned category and tags,

is persisted to the database. An immediate feed refresh is triggered so the new *Glimpse* appears for nearby viewers.

5.3 *Glimpse* Consumption

Published *Glimpses* are delivered to nearby viewers through a context-aware feed that adapts in real time to the viewer’s physical surroundings (**D1**, **D2**) (Figure 5.3).

5.3.1 Feed Display and Scrolling

The feed is rendered as a vertical tumbler on the left peripheral region of the OHMD, optimized for at-a-glance reading (**D3**). A sliding window of three *Glimpses* is shown simultaneously: the center item serves as the focal *Glimpse*, while the items above and below are visually deemphasized with a fade-out treatment to provide preview and continuity without competing for attention (**D3**). When idle, the list auto-scrolls by one position every 5 seconds, enabling lightweight scanning without sustained manual input during on-the-go use (**D4**) (Figure 5.3). Users can also scroll manually via the up/down buttons on the ring mouse controller. Each *Glimpse* displays a compact like indicator (icon + aggregated count), and viewers can double-click the left button to toggle a ”like” as a lightweight reaction cue (**D5**). Long-pressing the left button expands the currently focused *Glimpse* to reveal the full FPV photo along with situational metadata, the distance from the viewer’s current location (in meters) and the posting time, to aid in-place interpretation (Figure 5.4).

5.3.2 Two-Stage Context-Aware Filtering

The system employs a continuous, two-stage pipeline to determine which *Glimpses* to display, triggered by changes in the viewer’s visual environment.

In the Stage 1 (Visual Context Change Detection), a background thread continuously captures world-camera frames at approximately 3.3 Hz (one frame every 0.3s) and maintains a sliding window of the most recent 16 image embeddings,



Figure 5.3: The *Glimpse* consumption feed rendered on the left peripheral region of the OHMD. A sliding window of up to three *Glimpses* is displayed simultaneously: the center item serves as the focal *Glimpse*, while the items above and below are visually deemphasized with a fade-out treatment. Each *Glimpse* shows its cutout sticker, refined caption, and an aggregated “like” count. The feed auto-scrolls every 5 seconds and can also be navigated manually via the ring mouse controller. The three panels, from left to right, illustrate successive scroll states as the feed advances downward.



Figure 5.4: The expanded *Glimpse* detail view. Long-pressing the left button on the ring mouse controller expands the currently focused *Glimpse* to reveal the full first-person view photo with the highlighted region, alongside situational metadata including the distance from the viewer’s current location and the elapsed time since posting. This expanded view provides additional spatial and temporal context to aid in-place interpretation of the shared moment.

spanning roughly 5 seconds of observation. Each frame is encoded into a feature vector by MobileNetV3-Small, a lightweight convolutional neural network (CNN) specifically optimized for resource-constrained mobile and wearable devices [27]. The system computes the cosine similarity between each embedding in the window and a stored reference embedding representing the last stable visual context. If the fraction of dissimilar frames (cosine similarity < 0.6) reaches or exceeds 80 %, the system flags a scene change (e.g., transitioning from a street to a restaurant interior). The reference embedding is then updated to the current frame’s embedding and the

sliding window is cleared, resetting the detector for the next context shift.

In the Stage 2 (Distance and Semantic Filtering), the system executes a two-step content filter upon detecting a scene change. First, a distance filter retrieves all posts whose geographic coordinates lie within a 30 m radius of the viewer’s current GNSS position, computed via the Haversine formula. Second, an AI semantic filter sends the current world-camera frame together with a compact JSON summary of each nearby post’s identifier, category, and tags to the multimodal LLM. The model identifies the dominant scene type visible in the frame (e.g., park/greenery, food stall, street/building) and returns the subset of post identifiers whose category and tags are contextually relevant to the observed scene. Only these posts are retained in the feed.

5.3.3 Feed Update with Window Protection

When the filtered post set changes, the feed is updated incrementally to avoid disrupting the viewer’s current reading state. Posts currently visible within the three-item sliding window are protected: they are never removed mid-view, preventing abrupt content changes. Posts outside the window that are no longer in the filtered set are removed, and newly relevant posts are inserted immediately after the window boundary. If the window is exhausted (all three slots are empty after the user has scrolled past all content), the window position resets to the beginning of the updated feed. This peripheral placement, limited on-screen density, and auto-cycling behavior are consistent with prior heads-up computing practices [9, 8, 57, 66] and are intended to reduce interference with central vision during navigation (**D3**). Restricting social feedback to a single, low-effort “like” further matches the cognitive and motor constraints of ambulatory ring-mouse usage while still communicating engagement intensity in a glanceable form (**D3**, **D5**).

5.4 System Implementation

The *CityGaze* is implemented as a client–server application. The front-end user interface is connected to OHMDs (XREAL One¹), a Pupil Core eye tracker, a USB GNSS receiver (GlobalSat BU-353N5²) for precise geolocation, and a Bluetooth ring mouse controller (Sanwa Supply 400-MABT156BK2³) for lightweight interaction. It comprises a user-facing front end (client) and a back end (server) for intensive data processing.

The client is implemented with HTML, CSS, and JavaScript using ReactJS, and is used primarily to render and browse content generated by the back end. It communicates with the back-end server via RESTful HTTP requests, receiving processed outputs (e.g., selected crops, segmentation masks, and refined captions) in JSON format and displaying the resulting *Glimpses*.

The back-end server is developed in Python using the FastAPI framework and serves as the system hub: it ingests user inputs and sensor streams (GNSS information, FPV frames, and voice), orchestrates model calls, and manages database operations. Speech is transcribed live using the Azure Cognitive Services Speech SDK⁴. The server communicates with the Google Gemini model via API (gemini-3-pro-preview)⁵ to perform intention analysis, mask selection, and text refinement. The Segment-Anything Model (SAM 2.1 [49]) is deployed directly on the server to perform gaze-driven image segmentation during the *Glimpse* creation process.

The visual context monitoring pipeline described in the *Glimpse* Consumption section is implemented as a daemon thread on the back end. Frame embeddings are computed using a quantized MobileNetV3-Small model [27] via the MediaPipe Image Embedder API, which provides L2-normalized, quantized embeddings and

¹XREAL One: <https://next.xreal.com/one/>

²GlobalSat BU-353N5: <https://www.globalsat.com.tw/en/product-282242/USB-GNSS-Receiver-BU-353N5.html>

³Sanwa Supply 400-MABT156BK2: <https://direct.sanwa.co.jp/ItemPage/400-MABT156BK2/>

⁴Azure Cognitive Services Speech SDK: <https://learn.microsoft.com/en-us/azure/ai-services/speech-service/speech-sdk>

⁵Gemini API: <https://ai.google.dev/gemini-api/docs>

built-in cosine similarity computation. The sliding-window parameters (16 frames, similarity threshold = 0.6, change ratio = 80 %) were determined based on prior literature [9] and pilot testing.

To balance performance and portability, both the client and the back-end server run on a laptop with an Intel Core i7-13620H processor and an NVIDIA GeForce RTX 4060 GPU with CUDA support. The system architecture of *CityGaze* is shown in Figure 5.5.

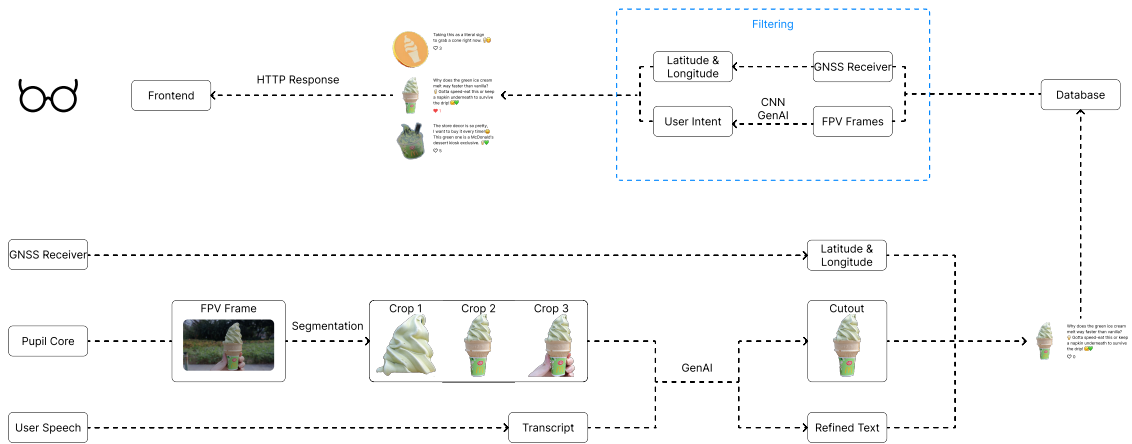


Figure 5.5: *CityGaze* system architecture. The front end client provides a lightweight display interface and visualizes *Glimpse* results produced by the back end. Sensor streams (GNSS information, FPV frames, and voice) are ingested and processed by the back end, which performs context-aware filtering (e.g., location/visual semantics), gaze-driven segmentation, and AI-powered content refinement. The pipeline illustrates both the creation flow, transforming raw FPV frames and gaze points into refined *Glimpses*, and the consumption flow, which continuously adapts the feed based on visual context changes.

Chapter 6

Exploratory Study

6.1 Study Overview

To understand how *CityGaze* is used in real-world scenarios, I conducted an exploratory study. This study was designed to explore how users engage with the context-aware generative heads-up system in dynamic on-the-go environments and to uncover their perceptions of *Glimpses* as a medium for connecting community members through shared physical contexts. This study specifically addresses the following two research questions:

- **RQ1:** How do participants engage with *CityGaze* to co-create and consume situated content (*Glimpse*) in dynamic physical environments?
- **RQ2:** How do participants perceive *Glimpse* as a socio-spatial medium for spontaneous expression and community connection?

6.2 Participants

12 participants (5 male and 7 female) were recruited via the institutional mailing list and convenience sampling. Their self-reported familiarity with smart glasses was generally low (Median = 2.0, IQR = 2.5; 1-7 Likert scale, 1 = no experience at all, 7 = very experienced), and their self-reported frequency of posting content on social media was generally high (Median = 6.0, IQR = 3.0; 1-7 Likert scale, 1 = never, 7

= multiple times a day). All participants self-reported having normal or corrected-to-normal vision. The study took approximately 90 minutes per deployment.

6.3 Apparatus

Participants wore the full *CityGaze* hardware setup described in Chapter 5. The display was provided by XREAL One glasses serving as the OHMD for the heads-up interface, paired with a Pupil Core headset for gaze-driven interaction. A Bluetooth ring mouse controller, held in the dominant hand, served as the input device for discrete interaction. The system ran on a backpack-mounted laptop that hosted both the client and server applications, providing portability while maintaining the necessary processing power for real-time AI generation. A mobile hotspot provided internet connectivity for API calls. In addition, a researcher followed each participant at a short distance and used a handheld camera to capture third-person view (TPV) video throughout the study session, documenting participants’ physical behaviors, body language, and interactions with the surrounding environment from an external perspective.

6.4 Study Site and Procedure

The 90-minute study was conducted along an approximately 800 m urban route encompassing high-density environments with four distinct scenes: open spaces (a public park and streets) and commercial venues (a restaurant and a convenience store) (Figure 6.1).



Figure 6.1: Overview of the designated study route. The site encompasses two types of environments with four distinct scenes: open spaces, including (A) a public park and (B) urban streets, and commercial venues, including (C) a restaurant and (D) a convenience store.

To simulate a realistic social awareness stream, I pre-populated the route with 36 *Glimpses* distributed across the four scenes. The design of these seed posts was guided by Naaman et al.’s content categorization framework for social awareness streams [48], with the categories simplified to better suit the study. This approach allowed us to examine consumption and engagement patterns across different physical and semantic contexts.

The study procedure consisted of three continuous phases. First, a 25-minute **onboarding and calibration** session involved hardware fitting, a 5-point eye-tracker calibration, and a tutorial on the "Look-and-Speak" workflow using the ring mouse. Next, participants completed a 45-minute **field walk** at their own pace. A researcher discreetly followed to record third-person perspective (TPV) video of their situated interactions. During the walk, participants performed two ad-libitum tasks: an *exploration task* to freely browse, expand, and react to pre-placed *Glimpses*, and a *content creation task* requiring them to author at least three new posts. Finally, a 20-minute **interview and questionnaire** phase concluded the session. Participants completed surveys on system usability and perceived task load, followed by a semi-structured interview exploring usability during mobility, engagement motivations, spatial perception, and the sense of social presence fostered by the asynchronous content.

6.5 Data Collection and Analysis

Four complementary data sources were collected: (1) timestamped system logs of user interactions, (2) first-person view (FPV) video captured from each participant’s perspective via the Pupil Core world camera, (3) third-person view (TPV) video recorded by the accompanying researcher using a handheld camera, and (4) semi-structured interview transcripts. The TPV footage provided an external observational record of participants’ bodily movements, navigation patterns, and social behaviors in context, complementing the egocentric perspective offered by FPV. These data were analyzed collectively using thematic analysis [5]. Interview tran-

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scripts, interaction logs, FPV recordings, and TPV footage were cross-examined to identify and iteratively refine recurring themes related to user engagement strategies and perceptions.

Chapter 7

Results

7.1 Overall Usability and Workload

Before presenting the qualitative findings, I briefly report two standardized metrics collected after each session. The system received a mean SUS [6] score of 75.00 ($SD = 12.92$), placing it in the *Good* to *Excellent* range on Bangor et al.’s adjective ratings [4]. NASA-RTLX [25, 24] scores indicated low workload for the use experience ($Mean = 22.71$, $SD = 17.97$). It is noted that ”mental demand” ($Mean = 14.58$, $SD = 19.36$), ”temporal demand” ($Mean = 18.33$, $SD = 27.66$), and ”frustration” ($Mean = 14.17$, $SD = 20.21$) were notably low, suggesting that the multimodal interaction flow was generally smooth with less attention needed (Table D.1). User-generated post categories are listed in Table E.1.

7.2 Engaging with *CityGaze*: Co-Creating and Consuming Situated Content (RQ1)

7.2.1 Commenting the Content via Simple Words

In the user study, participants tended to use simple words and concise expressions to convey their subjective feelings. Unlike traditional social media use on mobile phones or desktop/laptop computers, where users often carefully edit their wording and spend a long time preparing before posting, participants in the study frequently created content spontaneously. As a result, they sometimes became stuck and were

unsure how to continue. In such cases, *CityGaze* effectively supported participants. For example, P1 initially uttered only the fragment "who does" with the gaze fixed

on the  in a store, and the *CityGaze* pipeline refined the text into "Lemon tea popcorn?! Seriously, who does this? 🍋🍿😬", making the expression more complete and vivid while preserving the participant's original sentiment.

In addition, participants frequently relied on deictic expressions (e.g., "this") rather than explicitly naming the target objects. In such cases, *CityGaze* leveraged users' visual focus to infer the intended referent and incorporated the object's name into the generated text. For example, when P1 said "this is my favorite" while looking at a Vitasoy drink ,

CityGaze reformulated it as: "Nothing beats a classic Vitasoy. 🥤 Hands down my favorite! ✨".

In addition, automatic speech recognition errors occasionally occurred, especially in outdoor and public-space settings where the noise can be quite severe at times.

For example, when P2 looked at a billboard outside a restaurant , the speech-to-text system misrecognized "amazing" as "Amazon." and passed it to the GenAI. Grounded in the FPV visual context, *CityGaze* automatically corrected the error in the final output: "Taste of Southeast Asia looking amazing! 🍔🌶️ Definitely grabbing this."

To foster a stronger sense of connection with their audience, participants sometimes imagined a prospective viewer while creating content. For example, while

looking at , P9 said, "never forget to notice this tall building from the central of the [park]", which was transformed into a *Glimpse* with the text: "Never forget to take a moment and notice this massive skyscraper right in the center of it all! 🏙️👁️ Always look up. ✨". P7 also noted that "*I feel that the second-person approach lowers the barrier, creating a closer connection with the community and a more*

interactive feel".

7.2.2 Adapting Posture to Capture a *Glimpse*

While *CityGaze* facilitates mobile content capture, participants often adjusted their physical orientation to optimize the FPV angle. P7 highlighted this necessity, noting, "Overall, I think the interaction is quite smooth, but one thing is that I need to constantly adjust my posture in order to capture the content I want completely or accurately". To mitigate these framing challenges, participants adopted several compensatory strategies: leaning closer to targets (e.g., P1), manually lifting objects (e.g., P2), and pointing at objects to provide viewers with a secondary visual cue for confirmation when the view was expanded (e.g., P11) (Figure 7.1).

7.2.3 Leveraging Context-Awareness for Content Discovery

Since *CityGaze* was designed to be context-aware, participants expressed enthusiasm for its ability to help them discover hidden layers of information within their physical surroundings. A primary draw was the system's capacity for external references. P4 likened the experience to an "*AR version of Dianping*" (a review platform), noting that localized reviews for businesses like McDonald's provided helpful, real-time suggestions, while P1 found these cues valuable for overcoming "*trouble making choices*".

Beyond simple information retrieval, the system acted as a catalyst for exploration, P5 observed that surfacing related items simultaneously increased their "*interest in buying different things*", effectively expanding her consumption preferences while shopping. This discovery process was further enhanced by social feeds and visual stimuli. For example, P8 noted that the system brought attention to overlooked physical details like park murals, and P9 highlighted that the sticker-based design served as an intuitive trigger for exploration, allowing for quicker recognition of advertisements than traditional text. Interestingly, while users frequently "liked" content that captured their interest, this behavior was driven by personal resonance rather than the social influence of existing "like" counts.



Figure 7.1: Compensatory strategies for framing content. Top: P1 leans toward a refrigerated display to better capture a specific beverage container. Middle: P2 manually holds a snack bag within the FPV field of view to support accurate capture. Bottom: P11 points at a beverage bottle to provide a deictic cue, which allows viewers to confirm the focal object when expanding the associated digital sticker.

7.3 Perceiving *Glimpse* as a Socio-Spatial Medium (RQ2)

7.3.1 Fostering Digital Co-Presence and Connection

Beyond its functional utility, *CityGaze* served as an important social bridge, mitigating the loneliness often inherent in solo travel by fostering a profound sense of digital co-presence. Participants noted that discovering the *Glimpse* posted on the platform transformed solitary environments into shared spaces, with P4 describing the “*magical*” sense of connection that arises from seeing others’ footprints in the world. This feeling of belonging was specifically attributed to the human-centric nature of the content. Unlike “*template-based*” AI assistants, the user-generated content provides a sense of authentic companionship. For example, P11 remarked, “*this sense of interaction and communication made me feel as if someone was with me*”, while P5 highlighted that the “*abstract*” and unpredictable nature of human creativity offered a level of novelty and community that AI cannot replicate. Consequently, participants viewed the system not just as a way to learn about a location, but as a medium to uncover the human stories of those who traveled there before, “*I walked through the park not just because I wanted to learn about the park itself, but also because I wanted to hear stories about people [who had been there before]*” (P10). However, participants also identified that this social value is highly dependent on data density, noting that while the platform is well-designed, novel, and interesting, its full potential relies on a larger community to move beyond the study’s initial pre-populated data and create a truly live social ecosystem. “*it would be more interesting if more people shared content on this platform in the future*” (P5).

7.3.2 Balancing Attention Between the Physical and Digital Worlds

Participants noted that *CityGaze* achieves a harmonious balance between physical surroundings and digital social feeds, effectively moving beyond the screen-bound experience of traditional mobile devices. P4 compared this integration to "Danmaku" (bullet comments), where social layers and the primary visual field coexist seamlessly in a single viewport. Unlike the smartphone experience, which P8 described as being "*completely immersed in the world of the phone*" while ignoring the environment, *CityGaze* offers a more balanced "*50/50 split*" between the physical and digital worlds. P12 identified this transition from handheld screens to a seamless, integrated platform as the most significant breakthrough. Furthermore, participants emphasized the safety benefits of this heads-up interface. For example, P5 noted that while looking down at a phone is "*quite dangerous*" while on-the-go, the optical transparency of the glasses and the positioning of lightweight social content (i.e., *Glimpse*) in the peripheral vision allow for broader situational awareness, and this design facilitates users remaining attentive to their surroundings even while engaging with digital content.

7.3.3 Reducing Friction for Spontaneous Content Creation

Participants highlighted that the convenience of *CityGaze* directly fostered their creative expression. For instance, P3 rated the system highly compared to traditional sharing methods, noting that "*it can inspire my creativity anytime, anywhere*" simply by allowing them to look at an object and immediately input their thoughts. In addition, participants felt this seamless interaction actively promoted their desire to create content. P4 contrasted the "*instantaneous*" nature of *CityGaze* with traditional workflow of posting content on social media platforms, where the multi-step process of capturing a photo, thinking, and editing "*does not feel as instant and convenient*". P5 echoed this sentiment, observing that the inherent delay in smartphone-based sharing often diminishes a user's initial "*impulse and excitement*",

CHAPTER 7. RESULTS

whereas the immediate, gaze-based interaction of *CityGaze* captures thoughts in the moment.

Chapter 8

Discussion

8.1 Spatial Social Annotation as an Intent-Centered Paradigm

A defining characteristic of *CityGaze* is that it couples user-generated social content with the user’s real-time geolocation and visual context. Unlike screen-based social platforms where content exists independently of physical space [23], posts in *CityGaze* are created at, and retrieved near, specific geographic coordinates, and are further filtered by the semantic relevance of the viewer’s current scene. Posts are presented as a heads-up overlay rather than spatially anchored to specific surfaces, a design choice that preserves an unobstructed view of the environment, while the context-aware delivery helps keep content encounters tied to being physically present in the right place and looking at the right scene. In this regard, the experience resembles overlay commenting systems such as danmaku [60], where user reactions float over a shared visual medium without being pinned to specific pixels, except that *CityGaze* replaces the video timeline with geographic proximity and visual context as the organizing axes. While danmaku has been shown to create a sense of co-viewing and social presence among otherwise isolated viewers [60], *CityGaze* extends this notion from a temporal medium (video) to a spatial one (physical world), enabling asynchronous co-presence grounded in shared place and potential interest rather than shared playback time. Participants in the exploratory study echoed this effect: encountering others’ *Glimpses* at the same location reduced feelings of

solitude and created a sense of companionship despite the absence of synchronous interaction. A kindred dynamic is illustrated by Dear Data [43], in which two remarkable young women transformed daily observations into hand-drawn postcards exchanged asynchronously across continents, demonstrating that sharing personal, situated observations can forge deep companionship without synchronous contact. *CityGaze* transposes this dynamic to shared physical spaces. Each individual *Glimpse* capturing a fleeting personal observation grounded in place collectively becomes a social medium through which people connect over shared surroundings.

Crucially, the content produced through *CityGaze* is not reactive commentary on pre-existing media. It originates from an intentional act: the user selects real-world content through gaze, articulates a personal perspective through speech, and the system captures this as a first-person creative expression. The resulting post carries the trace of both the physical context in which it was created and the author’s situated attention at the moment of creation. This situatedness aligns with Harrison and Dourish’s foundational distinction between space and place, which suggests that while space provides the geometric opportunity for interaction, it is place imbued with social meaning and behavioral norms that frames how people actually engage with one another [23]. *CityGaze* operationalizes this distinction by transforming anonymous urban coordinates into socially annotated places through community contributions.

8.2 Context-Aware Delivery

Mainstream social platforms surface content through engagement-optimized algorithms that model a user’s historical preferences, often at the cost of serendipity and situational relevance [18]. *CityGaze* takes a fundamentally different approach by using content delivery governed by the user’s immediate physical context, geographic proximity, and real-time scene semantics. This design choice carries several implications observed in the user study. Participants noted that the feed felt meaningfully connected to their immediate physical surroundings. For example, *Glimpse* about

food appeared near restaurants, and nature-themed content surfaced in parks. This alignment between what users saw around them and what appeared in their feed was perceived not as algorithmic curation but as a natural extension of the environment itself. *CityGaze* imbued the space with social meaning [23]: the physical environment itself becomes the feed’s curatorial logic, grounding content relevance in shared place.

The context-driven model also appears to encourage a different mode of attention. In the exploratory study, several participants reported that browsing the *CityGaze* feed made them more aware of their physical surroundings, an experience they contrasted with the attention-absorbing nature of conventional social media. Where screen-based feeds compete with the environment for the user’s gaze, the heads-up presentation and context-matched content direct attention back toward the environment. This attention-enhancing quality, rather than attention-capturing, may represent a healthier relationship between social media consumption and real-world engagement. Prior work on digital well-being has identified excessive screen time and attention capture as primary concerns with smartphone-based social media [47, 2, 34, 38]. The design of *CityGaze* encourages a shift in the social interaction paradigm from passive scrolling toward situated, intentional engagement, where both content creation and consumption are grounded in the user’s immediate physical experience.

8.3 Human-AI Co-Creation and Authorship

The post creation pipeline in *CityGaze* is fundamentally a co-creative process. The user supplies intent by directing their gaze toward an object and speaking about it, while the AI shapes that intent into a publishable post by refining the spoken text and cropping a visual highlight from the scene. The final post (i.e., *Glimpse*) is a joint product that would not exist without both contributions. This human-AI collaborative pattern aligns with previous work in which humans initiate the content with AI for further improvement centered on human intent [65, 52]. In *CityGaze*,

the user’s gaze and speech serve precisely this steering function, grounding the AI’s generative freedom in situated, first-person intent. This framing aligns with findings from the formative study (Chapter 4), where creators reported substantially stronger intent alignment in the human-AI co-creation workflow than when the AI provided comments only.

Participants’ behavior during the exploratory study illustrates how this co-creation workflow enhances perceived social connection. Participants shaped their spoken commentary for an imagined future audience; they also reported reduced feelings of loneliness during solo exploration and a sense of connection when they read the *Glimpse* designed using this approach, aligning with findings that composing directed expressions for others enhances social connectedness [7].

8.4 Limitations and Future Work

8.4.1 Segmentation and Multi-layer Object Detection

To balance processing speed with cropping stability, *CityGaze* utilized multi-layer detection features from SAM triggered by a single gaze point. While effective in most scenarios, this approach struggles when an object contains more than three distinct semantic visual layers. Additionally, to simplify interaction for on-the-move use cases, the system relies on single positive prompts without negative prompts. This occasionally leads to inconsistent results, particularly when attempting to capture multiple separated objects. As multimodal models and agentic workflows become more efficient, future implementations could transform user gaze into a sophisticated set of positive and negative prompts. Such a system could iteratively refine results in the backend and provide hierarchical layer candidates to better adapt to dynamic contexts.

8.4.2 Technical and Hardware Limitations

While modern OHMDs are increasingly lightweight, the current *CityGaze* prototype relies on an external eye tracker and a backpack computing unit. The added bulk and thermal output of this hardware can reduce user comfort and mobility during prolonged daily use. Future iterations may prioritize miniaturizing components and improving thermal management to achieve a standalone form factor resembling traditional eyewear. Additionally, single-point gaze prompting remains highly sensitive to minor angular deviations, which may decrease selection accuracy for small or distant targets. To address this, future research may explore a "telescopic" gaze function to assist with distant objects and integrate affective semantic models to foster more nuanced, intent-aligned human-AI interactions.

8.4.3 Large-scale Social Deployment

The exploratory study utilized pre-created content to standardize the experience for all participants. However, this controlled environment may not fully reflect the complexities of large-scale social use. Real-world deployment involves intricate challenges, including recommendation algorithms and robust content moderation. While the current filtering mechanism may be sufficient for exploration in the early stage, a global deployment with millions of concurrent users would require a more sophisticated architecture to manage data privacy and social dynamics.

8.5 Vision

I envision *CityGaze* as a social interaction paradigm for a heads-up, intent-centered society. As OHMDs continue to evolve toward lightweight, always-on form factors [66], *CityGaze* could serve as the foundation for a new class of social platforms in which content is not consumed in isolation from the world but is encountered within it. It is worth further exploring such a lightweight social interaction paradigm with greater physical surrounding awareness, bridging the digital and physical worlds in service of richer, more situated social experiences.

Chapter 9

Conclusion

This thesis presents *CityGaze*, a context-aware social media platform for OHMDs that enables situated content creation and consumption on the move. The system contributes a novel "Look-and-Speak" interaction paradigm, distilling the authoring process down to a simple glance and spoken remark. Furthermore, it introduces a human-AI co-creation pipeline that pairs user gaze and speech with AI-driven cropping and captioning to generate lightweight social posts (i.e., *Glimpses*). This is supported by a two-stage, context-aware delivery mechanism that surfaces content based on both geographic proximity and visual-semantic relevance.

Empirical findings from an exploratory study provide strong support for this design. Quantitative metrics indicated good-to-excellent overall usability and low mental and temporal demands, suggesting a smooth, unobtrusive interaction flow. Qualitatively, *CityGaze* facilitated spontaneous content creation, with the AI pipeline effectively transforming simple words and deictic expressions into rich, contextually grounded posts. In addition, the study revealed that participants perceived *Glimpse* as a powerful socio-spatial medium that mitigates the loneliness of solo travel by fostering a profound sense of digital co-presence. Crucially, participants reported a balanced attention, maintaining high situational awareness, a contrast to the screen-bound immersion of conventional mobile phones.

Ultimately, this work positions *CityGaze* within an intent-centered model of social computing. As head-mounted displays continue to evolve toward lightweight, always-on form factors, the design principles established in this work offer a foundational blueprint for future platforms, shifting the paradigm from content consumed in

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isolation to authentic human stories discovered seamlessly within the physical world.

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Appendix A

Monthly Logs

A.1 October 2025

During this month, I designed and conducted a formative study with 12 travel enthusiasts to gain insights into how they perceive their travel experiences while using traditional mobile phones to consume social media content. These insights will inform the subsequent system design. Here are some of the main activities and progress made so far:

1. Designed the interview protocol of the formative study.
2. Recruited participants through targeted outreach and snowball sampling within the university community.
3. Qualitatively coded the interview data to identify key themes and design goals.
4. Designed the basic system functions.
5. Tested the prototype using smart glasses, configuring the color scheme and layout of the user interface.

A.2 November 2025

During this month, I tried different vision-language models and large language models to test their use in my project. In particular, I attempted to connect them together using some dummy components to test their usability and to refine the

implementation plan as needed. Here are some of the main activities and progress made so far:

1. Design dummy components for unimplemented parts for preliminary testing.
2. Test different models and analyze their limitations.
3. Implement the front-end user interface using ReactJS.
4. Test prompts for large language models and compare their effects.
5. Test the system using real smart glasses (XREAL ONE) in a real environment.
6. Refining the implementation and evaluation plans based on the feedback received during the presentation.

A.3 December 2025

During this month, I implemented the system based on the design plan formulated last month. In addition, I conducted unit tests for each module and debugged them accordingly. I consulted existing literature to gain insights that inform further iterations. I also connected the full pipeline, including the system integration with Microsoft Azure Speech-to-Text services, the Google Gemini API, and the Segment Anything Model, in order to test the entire procedure, as well as integration with a GPS receiver and Pupil Core eye tracking accessory products. Currently, the system runs smoothly. I added several fallback mechanisms to ensure it continues to work in special cases. Here are some of the main activities and progress made so far:

1. Implementing the complete system.
2. Conducting unit tests and analyzing potential issues.
3. Connecting the pipeline, including Microsoft Azure Speech-to-Text services, the Google Gemini API, the Segment Anything Model, Pupil Core eye tracking

accessory products, and a GPS receiver, and integrating them with the XREAL One Optical See-Through Display (OHMDs).

4. Formulating an iterative development plan to address existing issues and optimize the system.
5. Consulting the latest HCI literature, including work published in top venues such as ACM CHI, ACM CSCW, and ACM UIST, to gain insights for further system iteration and study/experiment design.

A.4 January 2026

During this month, I conducted a preliminary evaluation study using a prototype (rather than the full system). The study aims to assess the effectiveness of the proposed gaze-driven co-creation workflow and to identify refinements needed for in-the-wild deployment. Key activities and progress to date include:

1. Implemented the prototype system for the preliminary evaluation.
2. Recruited participants for the study.
3. Analyzed quantitative data using Wilcoxon signed-rank tests and coded qualitative audio data from retrospective think-aloud interviews through inductive thematic analysis.
4. Developed an iterative development plan to address current issues and optimize the system for in-the-wild deployment.
5. Recreated figures to improve clarity and presentation.

A.5 February 2026

During this month, I revisited the formative study, completed the full system implementation, and designed an exploratory study. Here are some of the main activities and progress made so far:

1. Completed the full system implementation.
2. Created thesis figures.
3. Conducted pilot tests for the user study.
4. Continued writing the thesis manuscript.
5. Reviewed additional literature and revised existing sections to improve clarity and presentation, incorporating self-proofreading and supervisor feedback.

A.6 March 2026

During this month, I completed an exploratory study and data analysis, and finished drafting and proofreading my thesis. Here are some of the main activities and progress made so far:

1. Completed an exploratory study with 12 participants.
2. Analyzed both quantitative and qualitative data and presented them in the thesis.
3. Wrote the results, discussion, and conclusion sections.
4. Wrote the acknowledgments to thank those who supported me during my undergraduate studies.
5. Revised the figures and text in the thesis.
6. Proofread the thesis before submission.

Appendix B

Pre-populated Content Design in the Exploratory Study

Table B.1: Categories and examples of the 36 pre-populated seed *Glimpses* used in the exploratory study. These categories are not mutually exclusive. A single post may naturally blend elements from multiple categories.

Category	Description	<i>n</i>	Example
Informational	Objective, factual content about a point of interest, such as historical background, ecological information, or practical tips	10	<i>"Straight outta Atlanta, 1886."</i>
Experiential	First-person sensory or emotional impressions of being there (describes the experience, no verdict or recommendation)	16	<i>"Obsessed with these blue stones."</i>
Evaluative	Subjective judgment or recommendation (good or bad, best or worst, worth it, preferences or comparisons)	22	<i>"Tomato flavor > Seaweed flavor."</i>
Conversational	Community-oriented messages designed to prompt social engagement, such as open-ended questions or calls for shared experiences	11	<i>"Anyone know why there's a random panda in the park?"</i>

Appendix C

Prompt Design for the System

This appendix documents all five prompt templates used by the AI service layer. All prompts are sent to the Google Gemini API alongside relevant image inputs.

C.1 Context Filter Prompt

Invoked after MobileNet detects a scene change. It receives the user's first-person view and a list of nearby posts (pre-filtered to ≤ 30 m), and returns which posts are semantically relevant.

```

Role: Context-aware content filter for a social platform.

Input:
- Image: [User's current first-person view from the world camera]
- Nearby Posts (already filtered to within 30 m):
{posts_json}

Task:
Look at the user's current visual context (the image) and decide which of the
nearby posts would be interesting and relevant to show right now.

Guidelines:
- Identify the primary scene type from the image (e.g., park/greenery,
  food stall/restaurant, street/building, indoor workspace).
- A post is relevant ONLY if its category/tags directly relate to the
  dominant visual elements in the scene.
  For example: trees/grass/flowers → nature/leisure posts;
  food stalls/vending machines/restaurants → food posts;
  buildings/signs/streets → architecture/urban/art posts.
- Be selective - exclude posts whose category does not match the scene.
  When in doubt, exclude the post rather than include it.
- Aim to return only posts from the one or two most relevant categories.
```

APPENDIX C. PROMPT DESIGN FOR THE SYSTEM

```
Output Requirement:
Return ONLY a valid JSON object. No markdown formatting.
{{
  "reason": "Brief one-sentence explanation of the user's current context.",
  "relevant_ids": ["id_1", "id_2", "id_3"]
}}
```

C.2 Speech Refinement Prompt

Refines raw speech-to-text into a social media caption. Receives the transcribed speech and the user's view with a gaze-point overlay.

```
Role: You are a witty social media editor.
Task: Lightly refine the user's spoken input into a short, punchy, human-like social media post.

Input Context:
1. User Speech: "{user_spoken_text}"
2. Visual Context: The attached image shows the user's view.
   - A RED DOT (semi-transparent) on the image marks the user's GAZE POINT.
   - Use the GAZE POINT to understand what the user is looking at and referring to (e.g., if they say "this" or "that").

Constraints:
1. Faithfulness: Capture the core meaning and intent of the user's speech.
2. Maximum Length: 200 English characters (including emojis).
3. Style: Casual, authentic, "human-like".
4. Emoji: Use naturally to express emotion.
5. Content Focus:
   - Remove filler words (e.g., "uh", "um", "you know") rigorously.
   - Fix misheard words based on the visual context.
   - Make it punchy and engaging.

Output Requirement:
Output ONLY the refined text.
```

C.3 Mask Selection Prompt

Selects the best segmentation mask (small, medium, or large) based on the user's speech intent.

```
Role: You are an expert intelligent photographer assistant.
```

APPENDIX C. PROMPT DESIGN FOR THE SYSTEM

Input:

- User Speech: "{user_speech}"
- Visual Context provided:
 1. Original Photo (The full scene)
 2. Cutout 0 (Candidate A - Smallest region)
 3. Cutout 1 (Candidate B - Medium region)
 4. Cutout 2 (Candidate C - Largest region)

Task:

Determine which Cutout (0, 1, or 2) is the **perfect visual match** for the object or concept described in the user's speech.

Selection Criteria:

1. **Semantic Match**: The cutout must contain the *entire* object mentioned, not just a part of it.
2. **Cleanliness**: The cutout should exclude irrelevant background clutter as much as possible.
3. **Visual Integrity**:
 - Avoid cutouts that crop off essential parts (e.g., a person without a head).
 - Avoid cutouts that are too zoomed out (e.g., the whole room when the user said "this cup").

Comparison Guide:

- **Cutout 0** is often too focused (might miss context). Use it only for small details.
- **Cutout 1** is typically the best balance for single objects.
- **Cutout 2** includes more surroundings. Use it if the user refers to a group, a large structure, or if Cutout 1 is too tight.

Output Requirement:

Return **ONLY** a valid JSON object:

```
{{  
  "reason": "Brief explanation of why this cutout is visually superior.",  
  "selected_index": 0, 1, or 2  
}}
```

C.4 Image-Only Baseline Prompt

Ablation baseline that generates a caption from visual input alone (no speech).

Role: You are a witty social media editor.

Task: Create a short, punchy, human-like social media post based on the attached image.

Input Context:

1. User Speech: [NONE] - You need to generate the caption from scratch.
2. Visual Context: The attached image shows the user's view.

APPENDIX C. PROMPT DESIGN FOR THE SYSTEM

- A **RED DOT** (semi-transparent) on the image marks the user's **GAZE POINT**.
- Use the **GAZE POINT** to understand what the user is looking at.

Constraints:

1. **Creativity**: Since there is no speech, you must creatively invent the caption. Feel free to be observational or funny.
2. **Maximum Length**: 200 English characters (including emojis).
3. **Style**: Casual, authentic, "human-like".
4. **Emoji**: Use naturally to express emotion.
5. **Content Focus**:
 - Identify the **OBJECT** indicated by the **RED DOT**.
 - Write the caption about that object.
 - Make it punchy and engaging.

Output Requirement:

Output **ONLY** the caption text.

C.5 Post Classification Prompt

Invoked after the user publishes a post. Assigns a category and tags for context-aware feed filtering.

Role: Content Classifier for a social media platform.

Task: Given a published post's caption and its image, classify the post with a content category and descriptive tags.

Input:

1. **Post Caption**: "{post_text}"
2. **Post Image**: The attached image is the post's sticker (a cutout of the subject).

Classification Rules:

1. **category**: A single lowercase word that best describes the post's subject.
Examples: "food", "nature", "shopping", "urban", "transport", "art", "animal", "architecture", "sport", "general".
Pick the most specific one that fits; use "general" only as a last resort.
2. **tags**: 3-6 lowercase tags describing the specific content of this post.
Tags should be based on what is **VISIBLE** in the image **AND** what the caption says.
Examples: ["sushi", "lunch", "japanese"], ["flower", "garden", "spring"], ["tram", "nostalgia", "city"].

Output Requirement:

Return **ONLY** a valid JSON object. No markdown formatting.

{{

APPENDIX C. PROMPT DESIGN FOR THE SYSTEM

```
"category": "single_word_category",  
"tags": ["tag1", "tag2", "tag3"]  
}}
```

Appendix D

NASA-RTLX Subscale Scores

Table D.1: NASA-RTLX subscale scores across all participants (N=12) in the exploratory study. Scale: 0-100.

Subscale	Mean	SD
Mental Demand	14.58	19.36
Physical Demand	27.92	30.49
Temporal Demand	18.33	27.66
Performance	35.83	34.17
Effort	25.42	23.59
Frustration	14.17	20.21
Overall (RTLX)	22.71	17.97

Appendix E

Categorization of User-Generated Posts

Table E.1: Distribution of user-generated posts by category across all participants (N=12) in the exploratory study.

Category	Count	Percentage (%)
Food	24	35.82
Shopping	11	16.42
Urban	9	13.43
General	6	8.96
Transport	5	7.46
Architecture	4	5.97
Nature	3	4.48
Animal	2	2.99
Toys	1	1.49
Art	1	1.49
Beverage	1	1.49
Total	67	100.00